Dummy Variables

School of Management and Economics
The Chinese University of Hong Kong, Shenzhen

The Linear Probability Model

- **A Binary dependent variable: the linear probability model**
- We studied how to incorporate **qualitative information** as independent variables in a multiple regression model
- We can also use regression to explain a **qualitative outcome**. For example,
  - we want to study the question of married women's labor force participation (in the labor force or out).
  - we want to know whether a young man is arrested for a crime during a certain period of time.
  - we want to know if an employee participates in a 401(k) pension plan.
- In these cases, the variable y we want to explain is a binary (zero-one) variable

The Linear Probability Model

- **A Binary dependent variable: the linear probability model**
- **Linear regression when the dependent variable is binary**

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$$

$$\Rightarrow \quad E(y|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k$$

The dependent variable only takes on the values 1 and 0

Probability that $y = 1$, probability of success

Probability that $y = 0$

$$E(y|\mathbf{x}) = 1 \cdot P(y = 1|\mathbf{x}) + 0 \cdot P(y = 0|\mathbf{x}) = P(y = 1|\mathbf{x})$$

$$\Rightarrow \quad \boxed{P(y = 1|\mathbf{x}) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k}$$

Linear probability model (LPM)

- The probability of success is linear in parameters β_j

The Linear Probability Model

- **Linear probability model (cont.)**

$$\beta_j = \partial P(y = 1|\mathbf{x})/\partial x_j$$

In the linear probability model, the coefficients describe the effect of the independent variables on the probability that y=1

- **The sample regression fuction**

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \ldots + \hat{\beta}_k x_k$$

Predicted probability of success

Predicted probability when each x_j is set to zero

Predicted change in the probability of success when x_k increases by one unit

The Linear Probability Model

- **Example: Labor force participation of married women**

  Data: MROZ.dta

=1 if in labor force, =0 otherwise

Non-wife income (in thousand dollars per year)

$$\widehat{inlf} = \underset{(.154)}{.586} - \underset{(.0014)}{.0034}\, nwifeinc + \underset{(.007)}{.038}\, educ + \underset{(.006)}{.039}\, exper$$

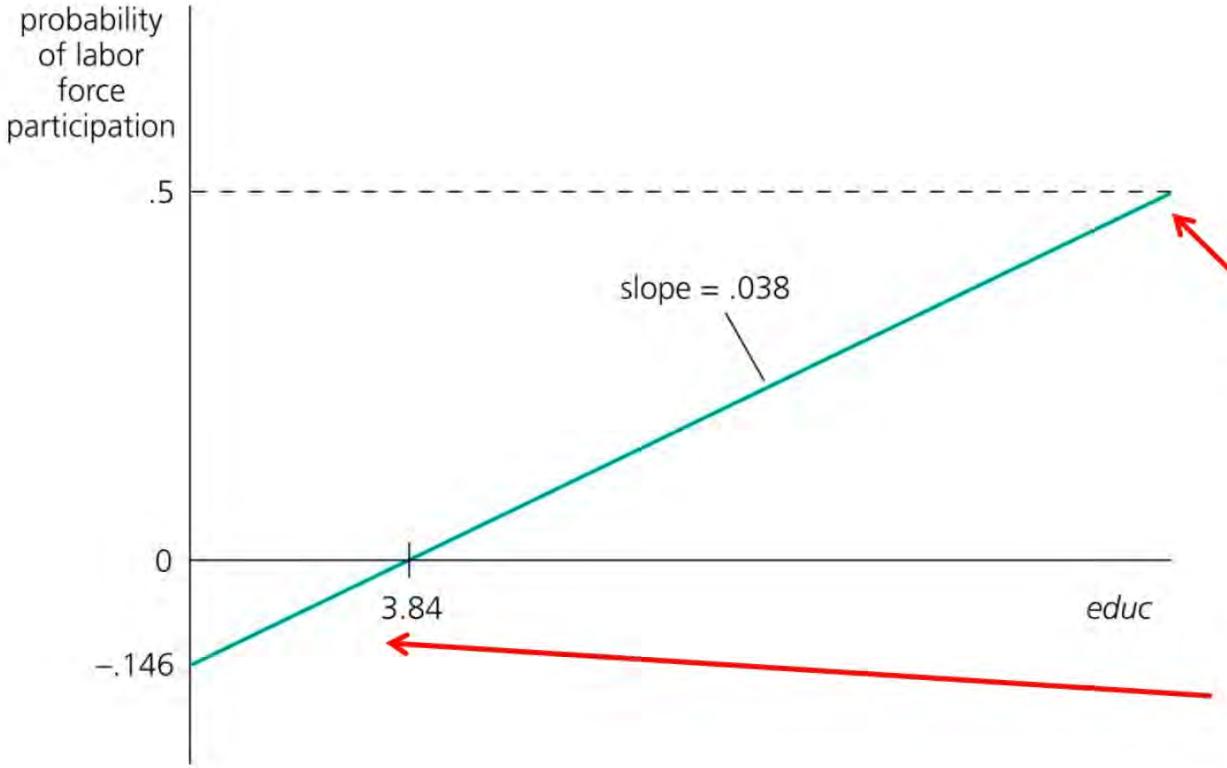
$$- \underset{(.00018)}{.00060}\, exper^2 - \underset{(.002)}{.016}\, age - \underset{(.034)}{.262}\, kidslt6$$

$$+ \underset{(.0132)}{.0130}\, kidsge6, \quad n = 753, R^2 = .264$$

If the number of kids under six years increases by one, the probability that the woman works falls by 26.2%

The Linear Probability Model

- **Example: Female labor participation of married women (cont.)**
- Relationship between the probability of labor force participation and years of eduction (with nwifeinc=50, exper=5, age=30, kindslt6=1, kidsge6=0)



The maximum level of education in the sample is educ=17. For the given case, this leads to a predicted probability to be in the labor force of about 50%.

Negative predicted probability but no problem because no woman in the sample has educ < 5.

The Linear Probability Model

- **Advantange of the linear probability model**
  - Easy estimation and interpretation
- **Disadvantages of the linear probability model**
- **Predicted probabilities may be larger than one or smaller than zero**
  - In fact, of the 753 women in the sample, 16 of the predicted values are less than zero, and 17 of the predicted values are greater than one
- **The estimated partial effects are constant thoughout all the possible values of independent variables**
  - For example, one more young child has a constant partial effects of 0.262, which implies that going from zero to four young children reduces the probability of working by:

$$\Delta\widehat{inlf} = .262(\Delta kidslt6) = .262(4) = 1.048$$

impossible!

The Linear Probability Model

- **Disadvantages of the linear probability model (cont.)**
- **The linear probability model must exhibit heteroskedasticity**

$$Var(y|\mathbf{x}) = P(y = 1|\mathbf{x})\,[1 - P(y = 1|\mathbf{x})]$$

Heteroskedasticity

- We can use this to find the conditional variance of the error term

$$\begin{aligned}
Var\,(u_i|X_{1i}, \cdots, X_{ki}) &= Var\,(Y_i - (\beta_0 + \beta_1 X_{1i} + \cdots \beta_k X_{ki})|\, X_{1i}, \cdots, X_{ki}) \\
&= Var\,(Y_i|\, X_{1i}, \cdots, X_{ki}) \\
&= Pr\,(Y_i = 1|\, X_{1i}, \cdots, X_{ki}) \times (1 - Pr\,(Y_i = 1|\, X_{1i}, \cdots, X_{ki})) \\
&= (\beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}) \times (1 - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_k X_{ki}) \\
&\neq \sigma_u^2
\end{aligned}$$

The Linear Probability Model

- **Goodness of fit in the linear probability model**
  - Besides R-square and adjusted R-squared, the LPM has other goodness of fit measures
- We can calculate the **percent correctly predicted,** by predicting a zero or one outcome depending on the estimated probabilities, i.e.

$$\begin{aligned} \text{Let } \tilde{y}_i &= 1 \text{ if } \hat{y}_i \geq .5 \\ &= 0 \text{ if } \hat{y}_i < .5 \end{aligned}$$

  then using the data on y_i and $\tilde{y}_i$ to obtain the proportion of overall correct predictions
- In Stata, it's very convenient to use comment "`tab`" to compare the data on two dummies variables

The Linear Probability Model

- **Example: Female labor participation of married women (cont.)**

1975		$\widehat{inlf}_i$ = 0	$\widehat{inlf}_i$ = 1	Total
$inlf_i$	0	203	122	325
	1	78	350	428
Total		281	472	753

Of the 324 women not in the labor force, we predict 203 correctly

Of the 428 women in the labor force, we predict 350 correctly

- Taken together, the overall percent correctly predicted is:

$$(203 + 350)/753 = .734, \text{ or } 73.4\%$$

Nonlinear probability models

- Probabilities cannot be less than 0 or greater than 1
- To address this problem we will consider nonlinear probability models

$$Pr(Y_i = 1) = G(Z)$$

$$\textit{with} \quad Z = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}$$

$$\textit{and} \quad 0 \leq G(Z) \leq 1$$

- We will consider 2 nonlinear functions

1. Probit

$$G(Z) = \Phi(Z)$$

2. Logit

$$G(Z) = \frac{1}{1 + e^{-Z}}$$

Nonlinear probability models

Probit regression models the probability that $Y = 1$

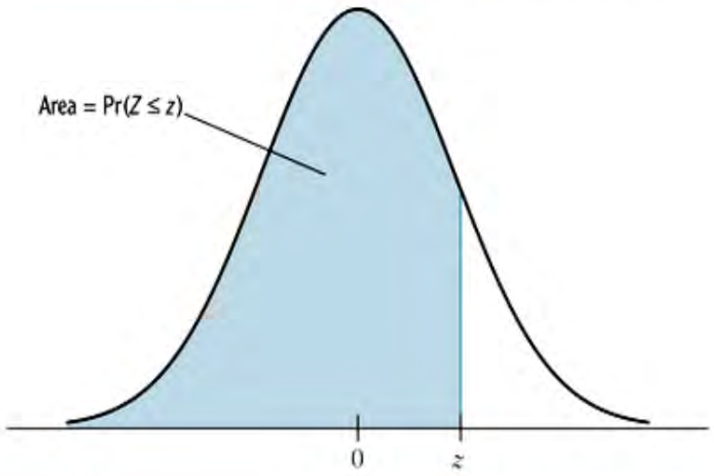
- Using the cumulative standard normal distribution function $\Phi(Z)$
- evaluated at $Z = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}$
- since $\Phi(z) = Pr(Z \leq z)$ we have that the predicted probabilities of the probit model are between 0 and 1

Example

- Suppose we have only 1 regressor and $Z = -2 + 3X_1$
- We want to know the probability that $Y = 1$ when $X_1 = 0.4$
- $z = -2 + 3 \cdot 0.4 = -0.8$
- $Pr(Y = 1) = Pr(Z \leq -0.8) = \Phi(-0.8)$

Nonlinear probability models

TABLE 1 The Cumulative Standard Normal Distribution Function, $\Phi(z) = \Pr(Z \leq z)$



	Second Decimal Value of z									
z	0	1	2	3	4	5	6	7	8	9
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121

$$Pr(Y = 1) = Pr(Z \leq -0.8) = \Phi(-0.8) = 0.2119$$

Nonlinear probability models

- Logit and Probit models are nonlinear in the coefficients $\beta_0, \beta_1, \cdots, \beta_k$
  - these models can't be estimated by OLS
- The method used to estimate logit and probit models is Maximum Likelihood Estimation (MLE).
- The MLE are the values of $(\widehat{\beta}_0, \widehat{\beta}_1, \cdots, \widehat{\beta}_k)$ that best describe the full distribution of the data.

Maximum likelihood estimation

- The **likelihood function** is the joint probability distribution of the data, treated as a function of the unknown coefficients.
- The **maximum likelihood estimator (MLE)** are the values of the coefficients that maximize the likelihood function.
- MLE's are the parameter values "most likely" to have produced the data.

Lets start with a special case: The MLE with no X

- We have n i.i.d. observations $Y_1, \ldots, Y_n$ on a binary dependent variable
- Y is a Bernoulli random variable
- There is only 1 unknown parameter to estimate:
  - The probability p that $Y = 1$,
  - which is also the mean of Y

Maximum likelihood estimation

Step 1: write down the likelihood function, the joint probability distribution of the data

- Y_i is a Bernoulli random variable we therefore have

$$Pr(Y_i = y) = Pr(Y_i = 1)^y \cdot (1 - Pr(Y_i = 1))^{1-y} = p^y(1-p)^{1-y}$$

  - $Pr(Y_i = 1) = p^1(1-p)^0 = p$
  - $Pr(Y_i = 0) = p^0(1-p)^1 = 1 - p$

- $Y_1, \ldots, Y_n$ are i.i.d, the joint probability distribution is therefore the product of the individual distributions

$$\begin{aligned} Pr(Y_1 = y_1, \ldots . Y_n = y_n) &= Pr(Y_1 = y_1) \times \ldots \times Pr(Y_n = y_n) \\ &= \left[p^{y_1}(1-p)^{1-y_1}\right] \times \ldots \times \left[p^{y_n}(1-p)^{1-y_n}\right] \\ &= p^{(y_1+y_2+\ldots+y_n)}\,(1-p)^{n-(y_1+y_2+\ldots+y_n)} \end{aligned}$$

Maximum likelihood estimation

We have the likelihood function:

$$f_{Bernouilli}(p;\ Y_1 = y_1, \ldots . Y_n = y_n) = p^{\sum y_i}\,(1-p)^{n-\sum y_i}$$

Step 2: Maximize the likelihood function w.r.t p

- Easier to maximize the logarithm of the likelihood function

$$ln\left(f_{Bernouilli}(p;\ Y_1 = y_1, \ldots . Y_n = y_n)\right) = \left(\sum_{i=1}^{n} y_i\right) \cdot ln\,(p) + \left(n - \sum_{i=1}^{n} y_i\right) ln\,(1-p)$$

- Since the logarithm is a strictly increasing function, maximizing the likelihood or the log likelihood will give the same estimator.

Maximum likelihood estimation

- Taking the derivative w.r.t p gives

$$\frac{d}{dp} ln\left(f_{Bernouilli}(p;\ Y_1 = y_1, \ldots . Y_n = y_n)\right) = \frac{\sum_{i=1}^{n} y_i}{p} - \frac{n - \sum_{i=1}^{n} y_i}{1-p}$$

- Setting to zero and rearranging gives

$$\begin{aligned} (1-p) \times \textstyle\sum_{i=1}^{n} y_i &= p \times (n - \textstyle\sum_{i=1}^{n} y_i) \\ \textstyle\sum_{i=1}^{n} y_i - p \textstyle\sum_{i=1}^{n} y_i &= n \cdot p - p \textstyle\sum_{i=1}^{n} y_i \\ \textstyle\sum_{i=1}^{n} y_i &= n \cdot p \end{aligned}$$

- Solving for p gives the MLE

$$\widehat{p}_{MLE} = \frac{1}{n} \sum_{i=1}^{n} y_i = \overline{Y}$$

Maximum likelihood estimation

Step 1: write down the likelihood function

$$\begin{aligned} Pr(Y_1 = y_1, \ldots . Y_n = y_n) &= Pr(Y_1 = y_1) \times \ldots \times Pr(Y_n = y_n) \\ &= \left[p_1^{y_1}(1-p_1)^{1-y_1}\right] \times \ldots \times \left[p_n^{y_n}(1-p_n)^{1-y_n}\right] \end{aligned}$$

- so far it is very similar as the case without explanatory variables except that p_i depends on $X_{1i}, \ldots, X_{ki}$

$$p_i = \Phi\left(X_{1i}, \ldots, X_{ki}\right) = \Phi\left(\beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}\right)$$

- substituting for p_i gives the likelihood function:

$$\left[\Phi\left(\beta_0 + \beta_1 X_{11} + \cdots + \beta_k X_{k1}\right)^{y_1}\left(1 - \Phi\left(\beta_0 + \beta_1 X_{11} + \cdots + \beta_k X_{k1}\right)\right)^{1-y_1}\right] \times \ldots$$

$$\times \left[\Phi\left(\beta_0 + \beta_1 X_{1n} + \cdots + \beta_k X_{kn}\right)^{y_n}\left(1 - \Phi\left(\beta_0 + \beta_1 X_{1n} + \cdots + \beta_k X_{kn}\right)\right)^{1-y_n}\right]$$

Maximum likelihood estimation

Also with obtaining the MLE of the probit model it is easier to take the logarithm of the likelihood function

Step 2: Maximize the log likelihood function

$$
\begin{aligned}
&ln\left[f_{probit}\left(\beta_0,\ldots,\beta_k;\ Y_1,\ldots,Y_n|\ X_{1i},\ldots,X_{ki},i=1,\ldots,n\right)\right] \\
&= \sum_{i=1}^{n} Y_i ln\left[\Phi\left(\beta_0+\beta_1 X_{1i}+\cdots+\beta_k X_{ki}\right)\right] \\
&\quad + \sum_{i=1}^{n}(1-Y_i) ln\left[1-\Phi\left(\beta_0+\beta_1 X_{1i}+\cdots+\beta_k X_{ki}\right)\right]
\end{aligned}
$$

w.r.t $\beta_0,\ldots,\beta_1$

- There is no simple formula for the probit MLE, the maximization must be done using numerical algorithm on a computer.

Maximum likelihood estimation

```
. probit deny pi_ratio

Iteration 0:   log likelihood =   -872.0853
Iteration 1:   log likelihood =  -832.02975
Iteration 2:   log likelihood =  -831.79239
Iteration 3:   log likelihood =  -831.79234

Probit regression                                 Number of obs   =       2380
                                                  LR chi2(1)      =      80.59
                                                  Prob > chi2     =     0.0000
Log likelihood = -831.79234                       Pseudo R2       =     0.0462
```

deny	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
pi_ratio	2.967907	.3591054	8.26	0.000	2.264073	3.67174
_cons	-2.194159	.12899	-17.01	0.000	-2.446974	-1.941343

- The estimated MLE coefficient on the payment to income ratio equals $\widehat{\beta}_1 = 2.97$.
- The estimated coefficient is positive and significantly different from 0 at a 1% significance level.
- How should we interpret $\widehat{\beta}_1$?

Maximum likelihood estimation

The estimate of β_1 in the probit model CANNOT be interpreted as the change in the probability that $Y_i = 1$ associated with a unit change in X_1 !!

- In general the effect on Y of a change in X is the expected change in Y resulting from the change in X
- Since Y is binary the expected change in Y is the change in the probability that $Y = 1$

In the probit model the predicted change the probability that the mortgage application is denied when the payment to income ratio increases from

0.10 to 0.20:

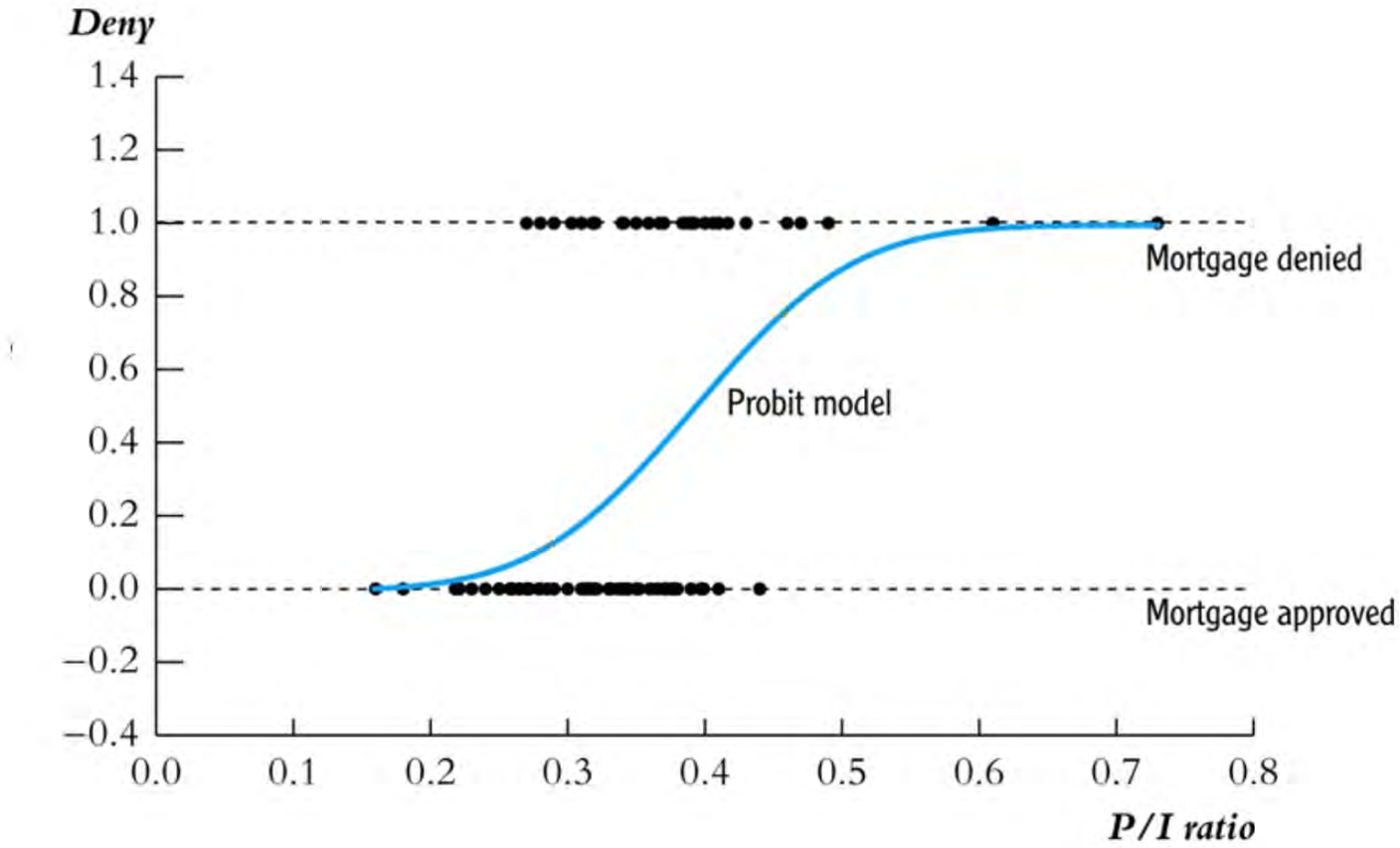
$$\widehat{\triangle Pr(Y_i = 1)} = \Phi(-2.19 + 2.97 \cdot 0.20) - \Phi(-2.19 + 2.97 \cdot 0.10) = 0.0495$$

0.30 to 0.40:

$$\widehat{\triangle Pr(Y_i = 1)} = \Phi(-2.19 + 2.97 \cdot 0.40) - \Phi(-2.19 + 2.97 \cdot 0.30) = 0.0619$$

Maximum likelihood estimation

Predicted values in the probit model:



- All predicted probabilities are between 0 and 1!

Maximum likelihood estimation

Table 1: Mortgage denial regression using the Boston HMDA Data

Dependent variable: *deny* = 1 if mortgage application is denied, = 0 if accepted			
regression model	LPM	Probit	Logit
black	0.177***	0.71***	1.27***
	(0.025)	(0.083)	(0.15)
P/I ratio	0.559***	2.74***	5.37***
	(0.089)	(0.44)	(0.96)
constant	-0.091***	-2.26***	-4.13***
	(0.029)	(0.16)	(0.35)
difference Pr(*deny*=1) between black and white applicant when *P/I ratio*=0.3	17.7%	15.8%	14.8%

Summary

- If Y_i is binary, then $E(Y_i|X_i) = Pr(Y_i = 1|X_i)$
- Three models:

1. linear probability model (linear multiple regression)
2. probit (cumulative standard normal distribution)
3. logit (cumulative standard logistic distribution)

- LPM, probit, logit all produce predicted probabilities
- Effect of $\triangle X$ is a change in conditional probability that $Y = 1$
- For logit and probit, this depends on the initial X
- Probit and logit are estimated via maximum likelihood
  - Coefficients are normally distributed for large n
  - Large-n hypothesis testing, conf. intervals is as usual